Trends in Available Commercially Zoned Lots in New Hanover County North Carolina

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March 24, 2026

1 Introduction

New Hanover County North Carolina has over 6000 commercially-zoned parcels. From 2000 to 2021 691 of these were sold, which is an average of 31.41 per year. Thus, 0.52% of the commercially zoned lots sold per year. The goal of this study is to establish trends in the availability of these lots. We seek to evaluate patterns in the number of lots listed and sold from 2000 to 2021. Of particular interest is whether availability has remained relatively constant, or if certain periods over the course of the study have had a higher (or lower) number of lots on the market. We also seek to determine what effect the recession (which started around 2008) had on the commercial real estate market.

The data comes from the John Hinant April 2021 New Hanover County vacant commercial land sale data set. For this study, only properties greater than one acre will be included. Industrial park locations will also be omitted. There were 252 parcels that met these criteria during the study period. It is important to note that parcels listed before July 1999 and after February 2021 are not included. Also, properties that sold before March 2000 or April 2021 are not included. The study is based only on the data set used, its conclusions on the number of properties on the market should not be viewed as exhaustive, as there may well have been properties, particularly early and late in the study, which were available for sale but not part of the data set. This is discussed further in the limitations section. Before looking at the pattern of available properties, it makes sense to look at some basic descriptive statistics:

Variable	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Sale Price	904.73 k	1.329 M	35 k	208.55 k	993.75 k	11.3 M
List Price	1.07 M	1.52 M	35.9 k	288.25 k	1.2895 M	13.3 M
Size(Acres)	8.9	24	1	1.7	7.4	246
Days on Market	663	767	0	200	799	4238
Price Diff.	-179.25 k	666.3	-5.75 M	-150 k	-8 k	5.4 M
Sale Price (per sqft)	5.2	8.4	0.067	1.4	6.2	107
List Price (per sqft)	6.4	11	0.069	1.9	7.1	145
Price Diff (per sqft)	-1.1	3.9	-38	-0.92	-0.055	6.3

Table 1: Descriptive Statistics

The average sale was slightly over $900,000, at $904,726. The most expensive parcel sold for $11,300,000, while the lowest price sale came in at $35,000. 50% of sales were between $208,548 and $993,750. The 90th percentile for the sale price was $1,987,500, while the 95th percentile was roughly $3,300,000. This indicates that the vast majority of sales were for less than $3,000,000, with a few very large outliers. There were ten sales over $4,000,000, and four over $7,000,000. Of these, there were none between $5,000,000 and $7,000,000, indicating that sales prices essentially topped out at $5,000,000, with four outliers that exceeded $7,000,000. The list prices ranged between $35,900 and $13,300,000, while 50% of the listings were priced between $288,250 and $1,289,500. Of the study parcels, 95% were listed for less than $3,570,000. Thus, the vast majority of properties both listed and sold for less than $3,500,000.

In addition to observing the prices for the properties listed and sold, the differences between the list price and the sale price of each individual property were examined. It is helpful to know if properties are selling for close to, or above, their listing price or if they are selling for significantly less than what they are listed for. The Price Diff variable in the above table is the difference between initial listing price and final sale price. It does not factor in any reductions in the listing price over the time the property remains on the market; only the initial listing and the final sale price are used. The middle 50% range for price difference is -$150,000 to -$8,000, so most of the properties sold below the listing price. The median price difference (which is not found in the above table) is -$40,000. It can be concluded that most of the properties sold for between the listing price and about $200,000 less than they were listed for. The mean difference was about -$180,000, but this is heavily influenced by several outliers. In particular, a property sold for about $5,750,000 below asking. Although this is somewhat balanced by another property that sold for $5,400,000 over listing, there were far more properties which sold for well below asking than well above, which accounts for the average price difference being about $180,000 below asking.

The average size of a parcel sold was 8.9 acres, while 50% of the parcels were between 1.7 and 7.4 acres. That the average parcel was larger than the 75th percentile for size is an indication of very large parcels skewing the average. The largest parcel sold was 246 acres, while the smallest was one acre, the minimum size included in the study. There are seven parcels larger than 50 acres, with three parcels between 50 and 70 acres, two being exactly 101 acres, one being 236 acres, and one being 246 acres. The 95th percentile for size was about 25 acres. Obviously, there were several very large parcels in the study, but the vast majority were smaller, with 75% of the parcels coming in below 7.4 acres. In terms of time spent on the market, 50% of the parcels spent between 200 and 799 days from listing to sale. The average time on the market was 663 days, while the property that took the longest to move sold after 4,238 days. The 90th percentile was 1421 days, which is close to four years. Thus, about 25 properties took four years or longer to sell, although the majority (about 70%) of parcels sold in less than two years. Typically, parcels remained on the market for between 6 months and 27 months, although some did take quite a bit of time to sell.

All the characteristics in Table 1 are related to market availability. However, for this analysis, the focus is on the number of properties available over the course of the study. The relationship between the above variables and the number of listed parcels can be evaluated in the future.

2 Results

In order to examine market availability, the study was broken down into different time frames. The initial period was from the start of the study, which was 2000, through 2007. The next period was the start of the recession, roughly 2008, through the peak of availability, which occurred in 2016. The third section covered the period after the availability peak, which would be 2017 to the end of the study. It is important to note that the analysis is based only on the available data set and, as stated previously, should not be treated as a complete representation of the properties on the market at either the beginning or the end of the study. In particular, the number of sales at the beginning of the study is probably under-counted, as properties that sold during the study period but were listed before the study began would not be included. Similarly, There were probably properties listed late in the study, perhaps 2019 or later, which are also not included. This is further discussed in the limitations section. The table below shows the sales, listings and transactions (which are listings and sales combined) for the three time periods in the study.

Time Period	**Number of Sales**	**Number of Listings**	**Transactions (Listings+Sales)**
Before 2008	33	42	75
2008-2016	120	155	275
After 2016	99	55	154

Table 2: Table of listings, sales and total transactions by time period.

Time Period	Number of Sales	Number of Listings	Transactions (Listings+Sales)
Before 2008	33	42	75
2008 – 2016	120	155	275
After 2016	99	55	154

Table 3: Table of listings, sales and total transactions by time period.

From 2008 to 2016, the combined number of transactions was much higher compared to the period before 2008. There were 275 transactions from 2008 to 2016, while there were only 75 from the beginning of the study to 2008. Obviously, the market was much more active during the second period of the study. The first period of the study (which is essentially 8 years) had 75 total transactions, for a rate of 9.375 per year. The second period, which was 9 years, had 275 total transactions for a rate of 30.55556 per year. As has been previously stated, the early years of the study probably underestimated the number of transactions, so the disparity is probably not as large as it seems, but it is still very clear that the market was more active from 2008-2016 than before 2008. After 2016 there were 154 transactions, which works out to 30.8 per year. Table 3, which is in the first appendix, contains a year by year breakdown of the total transactions by year, under the column labeled Total Changes. It can be easily seen that the market was very active from 2012 to 2018, with transactions ranging from a low of 38 in 2013 to a high of 56 in 2016. In 2018 there were 45 transactions, but in 2019 a steep decline is seen, as there are only 32 transactions, followed by 20 in 2020 and 7 in 2021, although the data set only covers through April 2021, indicating that the rate of transactions in 2020 and 2021 were quite close. All of this tells us that the rate of activity was fairly level in the second and third periods, with activity rising through the second period before leveling and staying steady through the first couple of years of the third period. The last three years of the last period show a decline in activity, although that could be due to the limitations of the data set. What is clear is that the market was far more active in the second two study periods compared to the first, although the exact difference in activity level is difficult to measure given the limitations of the data set.

In addition to analyzing the amount of market activity, the type of activity was also examined. The first two columns of Table 1 give an idea of whether the market had more listed properties or more that were sold. Obviously, a market where more properties are listed than sold would be a market where availability would be increasing. During the first period of the study, 33 properties were sold, while 42 were listed. Thus, listings were roughly 25% higher than sales before 2008. From 2008 to 2016 there were 120 properties sold while 155 were listed. Once again, the listings were about 25% higher than the sales. Obviously, during the first two periods, the high rate of listings relative to sales led to a substantial increase in available properties on the market. Although the market was clearly much more active in the second period, with 275 transactions from 2008 to 2016 compared to 75 before 2008, the ratio of sales to listings in both periods was essentially the same. During the third period this changes substantially, with 99 sales compared to 55 listings, there were 80% more properties sold than properties listed. Part of this is due to the data set, as at the end of the study every property included would have sold while no new listed properties were included, but it is still very clear that while the first two periods were characterized by higher rates of listings than sales, the third period saw far more properties sell in comparison to the properties that entered the market. It is likely that listings are under-counted for the third period; however, it is still quite clear that properties were selling at a high rate and availability was being reduced at this time.

To get an idea of whether there was a geographical pattern to the time properties were sold, a map was constructed. For this map, the time period is coded by color and shape, and the sale price of the parcel is represented by the size of the point. The map covers New Hanover County. By looking at a map of properties sold, we can get an idea of whether certain regions tended to sell during specific time periods, or if there was no relation between when a property sold and its location. If we see points of a certain color clustered in one area, that would indicate that parcels in that region were selling most frequently during the given time period. We can also get a sense of the prices of the parcels at different times. It would be expected that parcels that sold later would be more expensive due to appreciation, possibly with a dip due to the recession. However, it is also possible that properties that sold early in the study might simply have been more valuable, and thus a lack of appreciation in price would be due to the types of property selling at different times rather than a flat (or declining) market. In this case, the map does not show a particular geographical pattern related to time period and price. That is, points of a particular shape and color are not clustered in a specific region, the colors and shapes (which represent the different time periods) appear to be randomly scattered over the county. The same can be said of price; the points in any particular region do not appear to be larger in comparison to the other sections of the map. Like color and shape, large or small points appear to be randomly scattered. The map below, Figure 1, shows the locations of the parcels, the time period, and their price.

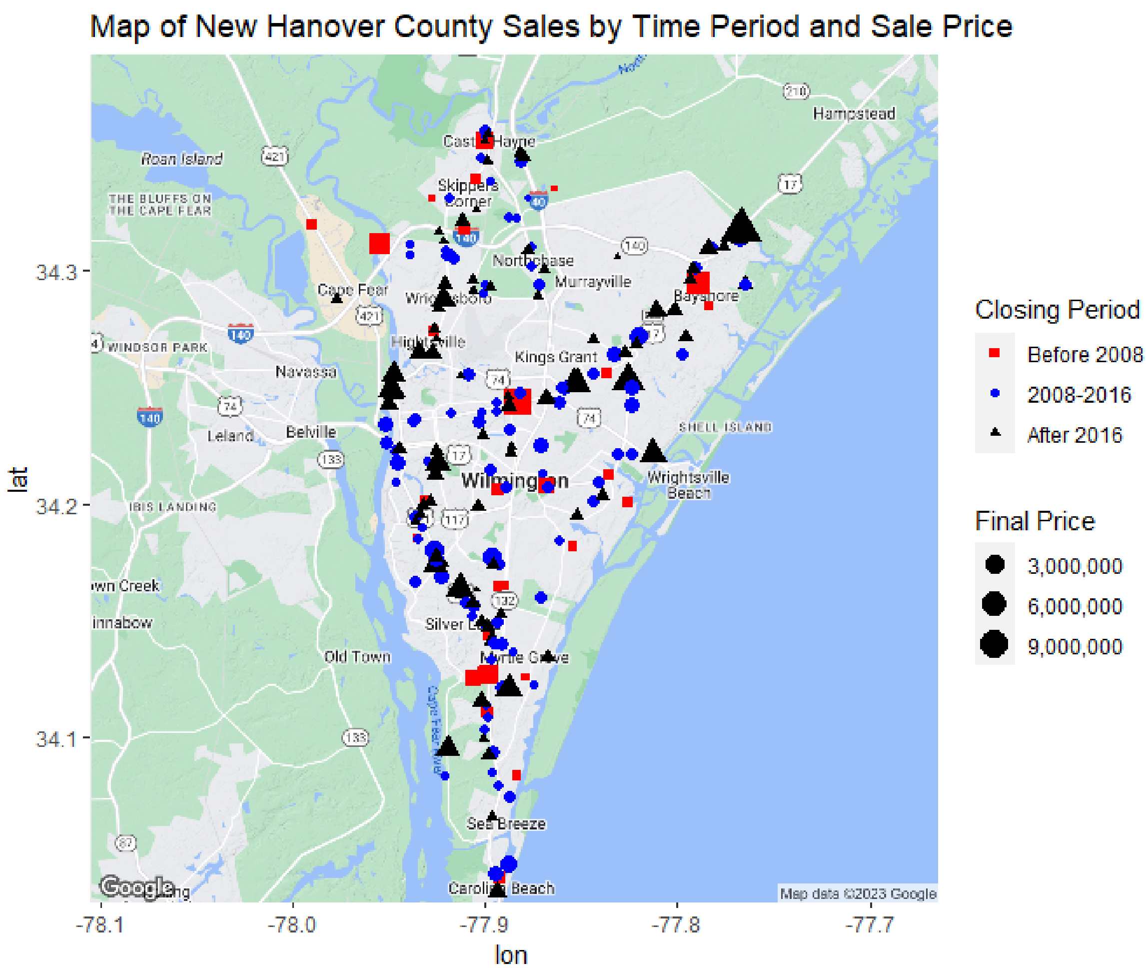

Figure 1: Map of sales by time period and price. There does not appear to be a geographical relation between price and time, as the points seem randomly scattered and to not have a pattern.

The red circular points on the map represent sales that occurred before 2008. There are far more blue and black points compared to the number of red points, reflecting the fact that far fewer properties were sold in the first period than in the next two, as can be seen in Table 2. There does not appear to be a geographical pattern by time period for sales. Parcels from the three time periods appear to be randomly scattered over the map, so if there is a relation between time sold and location, it is not obvious from the map. This could be studied in a more formally way in the future.

The main focus of this analysis is on the number of available properties of interest from 2000 through 2021. It makes sense to look at a plot of the raw number of available properties at each time point in order to get an idea of the pattern of availability over the study. Breaking sales and listings down into broad time periods is helpful, but it is also important to be more precise. We want to know just how many properties are on the market at each time point, not just how many are listed and sold over a multiyear period. Below is a plot of the number of parcels on the market over the course of the study:

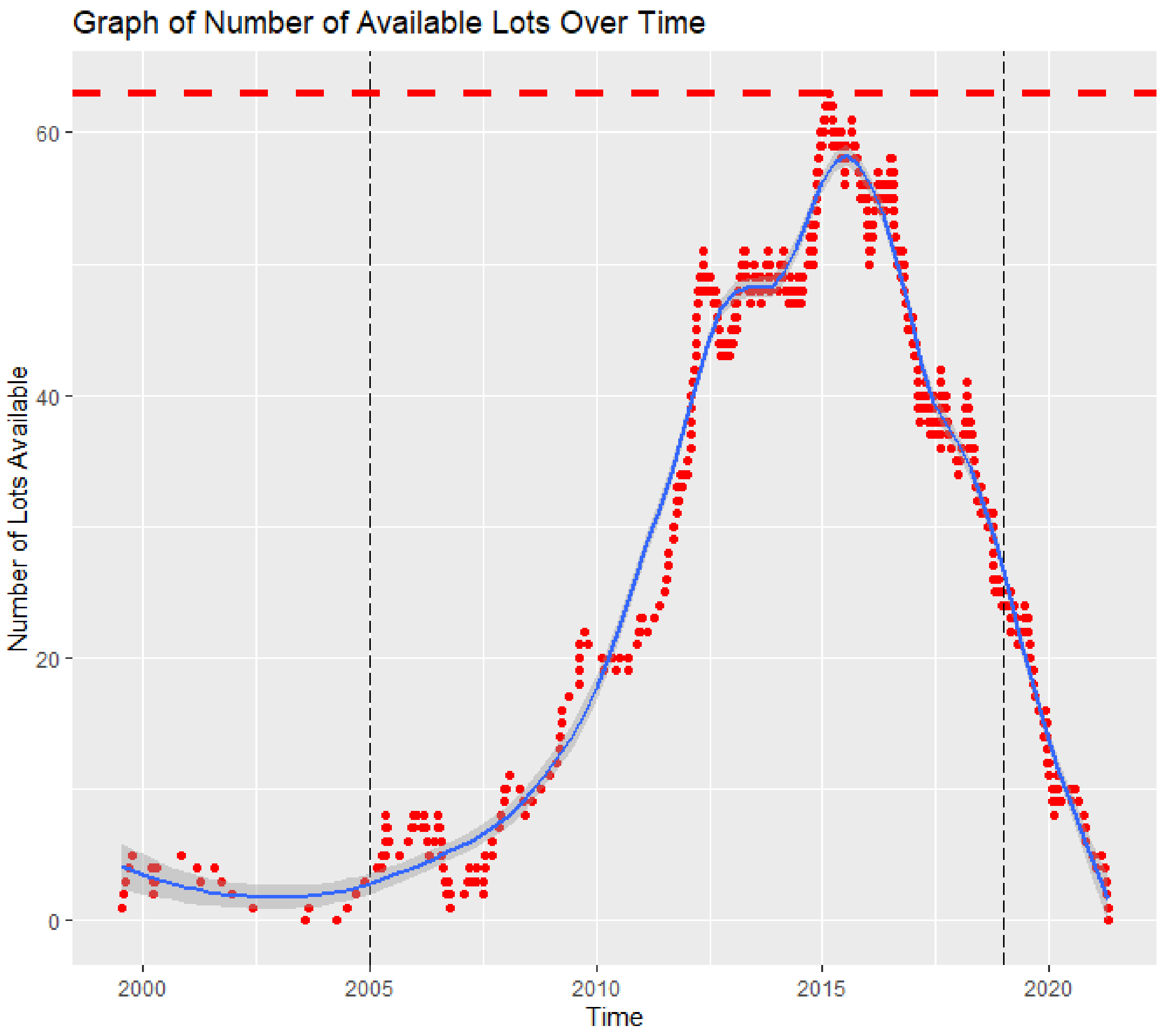

Figure 2: Plot of Number of Available Lots

The red points of Figure 2 (the above graph) represent the number of lots listed on the market at each given time point. The blue line is a smoother, which shows the general trend of the data over time, with the sharp edges rounded off. The type of smoother used is LOESS (Localized Estimated Scatterplot Smoothing) which is briefly explained in the appendix. Black vertical lines are included for the years 2005 and 2019. This was done to emphasize that the number of lots available is probably more accurate during the central period of the study, rather than at the edges. As has been previously stated, the early and late years of the study may not include all available properties, the study used the Hinant data set, and every property has a listing date and a sale date. Thus, the beginning of the study will automatically have zero properties available, while the end of the study will also not have any listings on the market because all properties included in the data set will have sold. The decision on which years to mark with vertical lines was not scientific; other years could have been chosen, the lines only emphasize that the count of available lots is likely to be more accurate over the central period of the study. The red dashed horizontal line at the top of the graph represents the peak of availability, which occurred in March 0 of 2015 with 63 parcels on the market.

The blue smoother gives a sense of the pattern of availability. Initially, a number of lots are listed, when they start to sell availability declines, which can be seen in the smoother having a slightly negative slope (which means decreasing in this context) from around 2000 to around 2003. The smoother than levels off, before starting to increase around 2004. From 2004 to 2009, the number of lots available is generally increasing, with the rate of increase becoming higher around 2009. This can be seen in how steep the smoother becomes after 2009, indicating that the number of properties on the market was sharply increasing from around 2009 to 2013. The number of properties on the market levels off around 2013, it is flat from around 2013 into 2014, before increasing again in late 2014. Availability peaks during 2015 and 2016, before starting to decline in the latter half of 2016. After 2016, it is easy to see that the number of properties on the market declines steadily and rapidly. Although the limitations in the data set make exact levels difficult, it is still very clear that there were substantially fewer properties on the market after 2018 compared to 2015 and early 2016, when availability was at its peak.

The number of parcels for sale at a given time is basically a reflection of the relationship between how many properties are being listed and how many are being sold. A decline in the count of available units can mean an increasing rate of sales with a level or decreasing (or possibly increasing) listing rate, or it could mean a level rate of sales with a declining listing rate. The raw count tells us only how many are for sale; it does not say anything about the listing and sales rates. A market where an increase in availability is the result of a decrease in the rate of purchases might be very different from a market where the increase is the result of a higher rate of listings. It makes sense to examine the sales and listings rates over the course of the study.

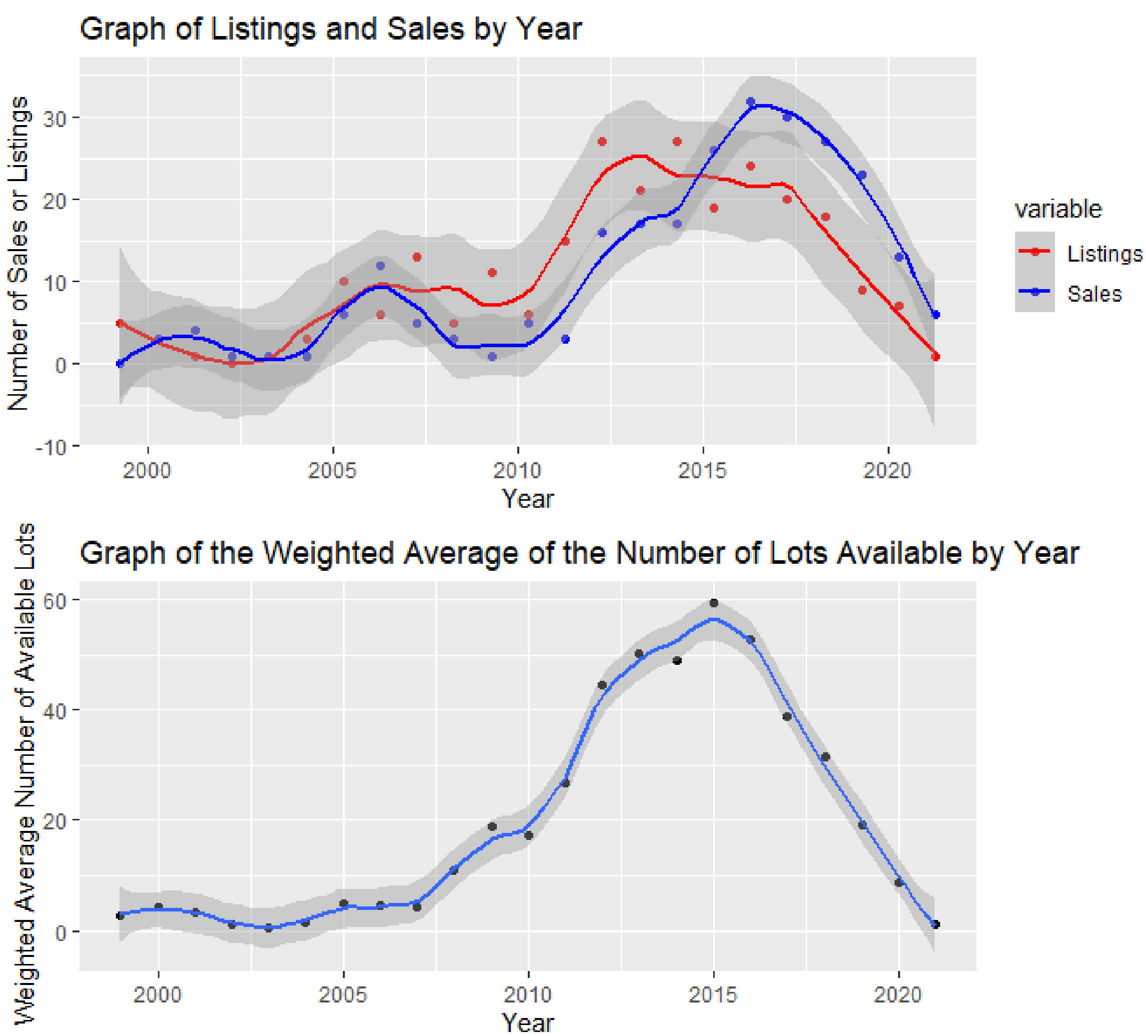

Figure 3: Top plot is of number of sales and listings by year. Bottom plot is weighted average of available properties by year.

The red points in the top plot of Figure 3 represent the total number of listings for the given year. The blue points on the same graph are the number of sales for each year. As in Figure 3, smoothers have been included to give a feel of the general pattern of sales and listings. The bottom graph of Figure 3 shows a weighted average of the number of properties available by year. Essentially, the more days the market has with a certain number of properties available, the heavier that number is weighted towards the average for that time period. If the time period is 30 days, and there are 50 properties available for 20 days and 30 for 10 days, when calculating the weighted average the method would assign more weight (in this case 20) to the number 50 than to the number 30, as there would be more time with 50 properties available than 30 properties for sale. The weighting scheme is explained more thoroughly in the methods section, which is in the appendix, but the weighted average gives an estimate of the number of properties on the market over the course of the year. A smoother is also included in the bottom plot. Table 3, which is in the first section of the appendix, contains the yearly weighted averages found in the bottom plot. If the bottom plot is compared to Figure 2, the shape of the graph is essentially the same. It can be easily seen that the listing and sales pattern in the top graph of Figure 3 is reflected in the available property pattern in the bottom graph. In the top plot, the blue and red lines are quite close through about 2006, which is consistent with the bottom plot, where the slope of the smoother is fairly flat through about 2007. This indicates that listings and sales were closely matched through around 2006, and properties were not sold or listed, at significantly different rates during this period. After 2006, the number of listings per year is fairly flat; it is higher in some years and lower in others, but the average number of listings for the period from 2005 to 2010 is 8.5, while the average number of listings from 2011 through 2016 is just over 22. This can easily be seen in the red line in the top graph, which is fairly flat from 2005 through 2010, but quite steep from 2011 to 2014 before leveling off again. In contrast, the sales plot (blue line and points) declines after 2005, before flattening to a much lower level around 2010. Thus, while listings were fairly level between 2005 and 2010, sales declined significantly.

The decline in sales paired with flat listings led to an increase in availability between 2007 and 2010, which can be easily seen in the bottom plot in Figure 3. After 2010, both sales and listings rates increase, which can be seen in the steep slopes of the red and blue lines after 2010. Availability increased after 2010 as even though more properties were selling, the rate of listings was still significantly higher than the rate of sales. The bottom graph shows the steady increase in property on the market from 2010 through 2016. In 2015, the rate of listings levels off, then starts to decline, which is clear from the red line in the top graph, while the rate of sales (blue line, top graph) continues to increase until around 2016, before declining sharply starting in 2018. The sales rate moves past the listing rate around 2015, which can be seen in the top graph where the blue and red lines cross. The bottom graph reflects the sales/listings relation shown in the top graph: The peak of availability occurs during 2015, a point at which the increase in sales rate had not yet compensated for the higher rate of listings that had occurred over the last 7–8 years. After 2015 we see the high rate of sales leading to a decline in properties on the market, with the decline slight in 2016 and more precipitous from 2017 through 2019, when we see the steep negative slope in the bottom graph, a result of sales rates being significantly higher than listings rates. It is clear that sales and listings followed somewhat different patterns over the course of the study. The rates were similar for the first few years, followed by a decrease in sales while listings remained steady. The listings then increased dramatically, while sales followed suit a bit later. Although both rates were increasing during this time, the rate of listings was substantially higher than the rate of sales, leading to a spike in parcels on the market that peaked around 2015 or early 2016. Sales rates surpassed listing rates in 2105, which eventually led to a precipitous decline in availability beginning around 2017.

In addition to looking at listings and sales, it also makes sense to examine how active the market is. In this case, the activity is the number of listings and sales. For example, if the weighted average of the number of available properties for a given year is 50, all that tells us is the typical number of properties on the market in a given year. A weighted average of 50 with 30 sales and 30 listings in a given year would be a very different market from one where there were 5 listings and 5 sales, but the same weighted availability average. In the first scenario, the market is very active, even if the number of available parcels remains steady, as properties are being listed and sold at a high rate. In the latter case, there are very few listings and sales, and thus a much less active market. To distinguish between these situations, two metrics were examined: The first metric was total changes, which is simply the number of listings in a given year combined with the number of sales. The second is the difference, which is the total number of sales subtracted from the total number of listings. A high number of total changes indicates an active market. A high positive difference indicates a market where there are many more listings than sales. Obviously, in a market with a high positive difference availability would rise. A high negative difference would characterize a market with more sales than listings, which would result in a decline in availability. If the difference is close to zero, there would be little change in the available properties. The graph below charts these metrics for the time period:

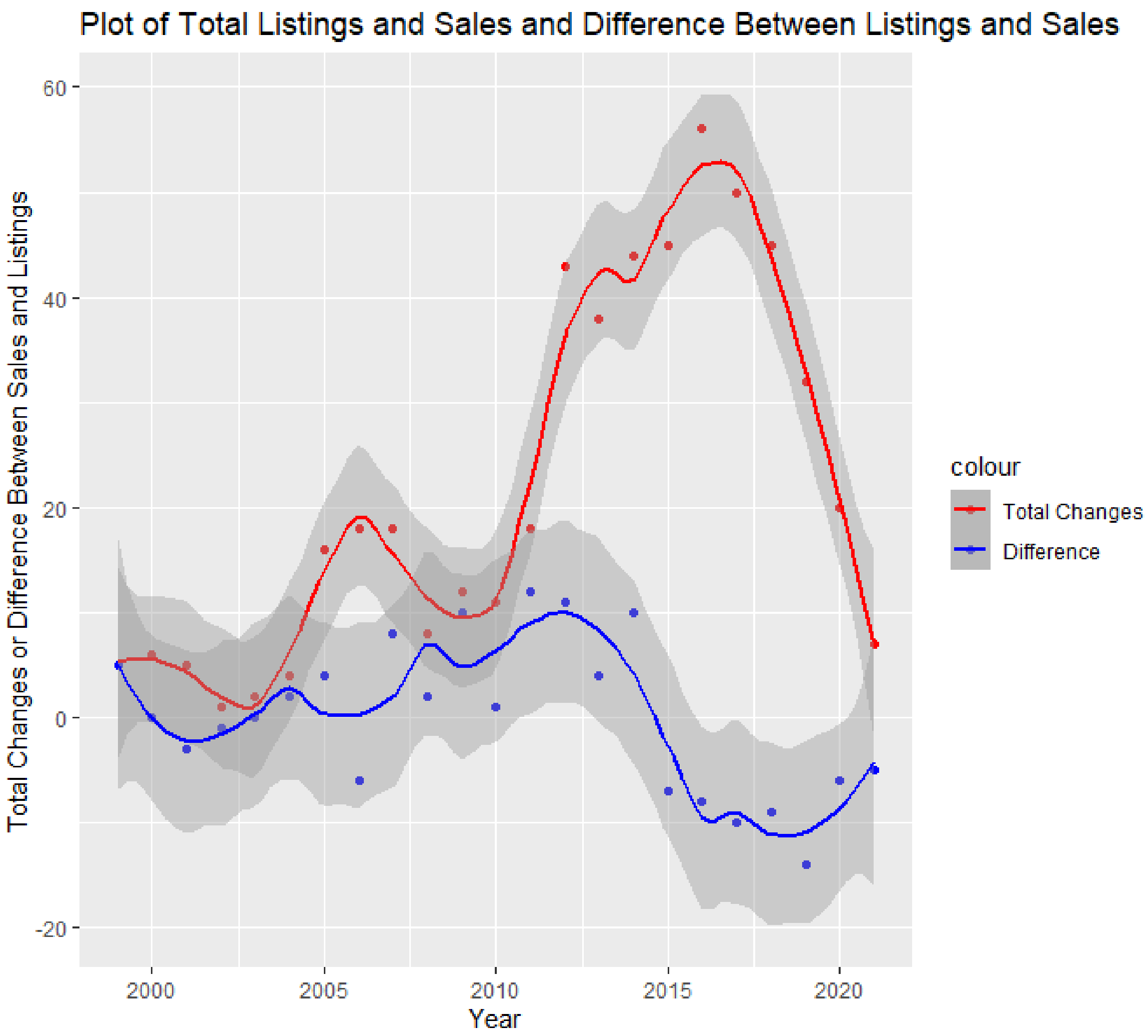

Figure 4: Plot of the total listings and sales for each year as well as the difference between sales and listings for each year

The red points in the plot above represent the total changes. After 2010, the market became much more active, which can easily be seen by the steep, red smoother between 2010 and 2016. Before 2010, the total changes did vary, but there was never a year with more than 20 total listings and sales. After 2010, it is easy to see the spike in changes that indicates a much more active market. Starting around 2014, which can be seen in the blue points and the blue smoother line, the difference between listings and sales dips sharply, from being positive (at +10) to -7 in 2015. This indicates a period where substantially more properties were sold than put on the market. Obviously, this steep decline in difference coincides with the post 2015 section of the top graph of Figure 3, where sales are substantially higher than listings. It also matches the steep decline in the weighted average of lots available after 2015, which can be seen in the bottom graph of Figure 3.

Overall, availability did fluctuate substantially during the study period, with the number of parcels on the market rising significantly after 2008, peaking around 2015, and then declining sharply after 2016. Although limitations in the data make it difficult to calculate exact parcel availability after 2019, it is still very clear that the number of available properties peaked in 2016 and declined dramatically afterward.

3 Limitations

The major limitation is the structure of the data, which has already been mentioned several times. Each property in the data set has both a listing date and a closing date. Thus, at the beginning of the study there will not be any properties on the market, as none have listed, and at the end of the study there will also not be any properties available, as they will all have sold. Because there is no information about properties that are not in the data set, the actual number of properties available at the beginning or end of the study is not accurate. Towards the center of the study the availability is likely very accurate, but it is not reasonable to assume that after the last property sold in 2021 there was nothing on the market. Likewise, it is ridiculous to assume that before the first property in the study was listed, in July 1999, there were no qualifying parcels on the market.

4 Appendix

4.1 Availability Table

Year	***Weighted Average Available Properties***	***Listings***	***Sales***	***Total Changes***	***Difference***	***Median Sale Price***	***Median Price Difference***
1999	2.825	5	0	5	5	NA	NA
2000	4.224	3	3	6	0	165000	-27500
2001	3.463	1	4	5	-3	326000	-2000
2002	1.151	0	1	1	-1	275000	-55400
2003	0.638	1	1	2	0	250000	-149000
2004	1.418	3	1	4	2	350000	-20000
2005	4.885	10	6	16	4	69500	0
2006	4.477	6	12	18	-6	1800000	-31950
2007	4.334	13	5	18	8	182500	-12500
2008	11.003	5	3	8	2	875000	-20000
2009	18.718	11	1	12	10	6e+05	-680000
2010	17.332	6	5	11	1	770000	-5000
2011	26.693	15	3	18	12	884000	-1500000
2012	44.527	27	16	43	11	787500	-111950
2013	50.107	21	17	38	4	4e+05	-92000
2014	48.847	27	17	44	10	487500	-115000
2015	59.310	19	26	45	-7	397312.5	-50202.5
2016	52.803	24	32	56	-8	422500	-29900
2017	38.729	20	30	50	-10	751875	-75000
2018	31.619	18	27	45	-9	625000	-75000
2019	19.123	9	23	32	-14	549000	-33250
2020	8.899	7	13	20	-6	179500	-10000
2021	1.090	1	6	7	-5	544000	-43000

Table 4: Table of metrics for each year. This table has weighted average of available properties by year, total listings by year, total sales by year, total changes (listings + sales) per year, the difference between total listings and total sales by year, the median sale price for each year, and the median difference between list price and sale price for by year.

Table 3 contains the weighted average number of properties available by year. It also contains a number of the other yearly metrics used in the report, namely listings and sales per year, as well as total changes. In addition, it contains the median sale price and the median price difference for each year. The price difference is the difference between the sale price and the listing price for each property. A negative price difference indicates that the listing price was higher than the sale price.

4.2 Methods

1. **Weighted Average**

   For this report, no high level statistical methods were used. However, weighted averages were used in several places. A weighted average is of the following form:

   $$Z = \frac{\sum_i^n \delta_i * X_i}{\sum_i^n \delta_i}$$

   In the above equation, Z represents the weighted average, the δ_i values are the weights, and the X_i values are the observations. Thus, each observation X_i is assigned a weight δ_i, then each observation is multiplied by its weight, and these are all added to form the numerator. The denominator is simply the sum of the weights.

   For this study, this methodology was used to develop the weight average of availability by year. For each year, each level of availability was the X_i value in the above equation, and the number of days at the level of availability was the δ_i. To look at a simple example, if this were to be done for a month when there were 40 lots available for 25 days and 50 lots available for 5 days, we would have $X_1 = 40$, $\delta_1 = 25$, $X_2 = 50$ and $\delta_2 = 5$. If these values are plugged into the above equation, we get:

   $$Z = \frac{40 * 25 + 50 * 5}{25 + 5} = 41.667$$

   If the two availability numbers had been averaged with equal weights, that would have yielded a metric which does not take into account that fact that far more time was spent with 40 lots available than with 50. The above weighted average for the 30 days is 41.667, which reflects the fact that for most of the time period, there were 40 lots for sale.

   In this study the weighted average methodology described above is used to generate the data for the bottom graph in Figure 3. For this plot a weighted average of the number of lots available was done for each year.

2. **Smoothing Method**

   Most of the graphs in this report contain smoothers. The purpose of a smoother is to give an idea of the trend of general pattern, or trend, of the data. A smoother forms a line, likely a curving line, through the points which follows the data trend, essentially 'smoothing' out the rough edges.

   For the graphs in this project, the smoothing method used was LOESS, which stands for locally estimated scatterplot smoothing. LOESS is a form of local regression. The general procedure for local regression is as follows:

   (a) First, a group of pairs of points is needed. These would be of the form:

   $$(X_1, Y_1), (X_2, Y_2), ...(X_n, Y_n)$$

   (b) Form a neighborhood of points around the given point, which we will call X_0. The size of this neighborhood can be determined in several ways.

   (c) Assign a 'weight' to each one of the points in the neighborhood, other than the point of interest. There are several different choices for the weight function. Typically, the idea is that points close to X_0 will have a higher weight than points far from it. In this way, points similar to the point of interest will have a stronger influence on the estimate.

(d) Perform a regression in order to form the specific linear model for the given point X_0. This will give us an estimate of Y_0 for each X_0. We will call this estimate $\hat{f}(X_0)$. This estimate will have the form:

$$\hat{f}(X_0) = \hat{\beta}_0(X_0) + \hat{\beta}_1(X_0) * X_0$$

The $\hat{\beta}_0(X_0)$ and $\hat{\beta}_1(X_0)$ in the above equation are the estimated regression coefficients for the point X_0. That is, they are the results of performing a weighted regression analysis based on the points in the neighborhood of the given value X_0. They are the results of solving the following:

$$\min_{\beta_0,\beta_1} \sum_{i}^{N} K_\lambda(X_0, X_1)[Y_i - \beta_0](X_0) - \beta_1(X_0)X_i]^2$$

Where $K_\lambda(X_0, X_1)$ is the weight of the point X_i among the neighborhood of points for X_0. The neighborhood in the above equation would be of size N. Thus, for a given point X_0, there would be N different weights, one for each of the points in the neighborhood.

There are many different formulas for determining the weights. Most attribute higher weights to points that are closer to X_0. For this graphs in this project the smoothers are generated using weights of the form:

$$K_\lambda(X_0, X_1) = D(\frac{|X_i - X_0|}{h_\lambda(X_0)})$$

Where $h_\lambda(X_))$ represents the width of the neighborhood, which can either be constant or variable. For the purpose of this study constant widths are used in smoothing. The D() in the equation above represents a distance function. There are many different distance functions; for this analysis, the function used is the *tri-cube* function which takes the following form:

$$D(t) = \begin{cases} (1 - |t|^3)^3 & if\ |t| \leq 1 \\ 0 & Otherwise \end{cases}$$

In the above function, t represents the distance from the point of interest. The larger the value of t, the lower the weight attached to that point. Thus, the method weights points that are close to our point of interest higher than points that are further away.

(e) Continue for all the given points. Thus, each point will have its own linear model with which to generate its estimated value. This estimate will be based on a weighted regression of the points in its neighborhood. We will then have a series of estimates of the y_i values, from $\hat{f}(X_1)$ to $\hat{f}(X_n)$, and the plotted line of these points will be the smoother line.

4.3 Other Questions

1. **Trend in property being offered on the market (in column: created date) over 20 years, including how many parcels were being advertised for sale at any time**

   This is the focus of this analysis.

2. **Rate/ number of land sales (closings) per year. This will reflect the 2007-2012 slowdown.**

   This is part of the analysis.

3. **Sales ranked by named road location over 20 years (How can this be done? It will be necessary to remove street numbers and select road addresses and rank them)**

   The address field has been broken down into separate columns for steet names and street addresses. This could potentially be used to model this question, although it is not part of the analysis.

4. **Tracking the size of parcels sold each year**

   This has been done. It is touched on in the Introduction. It can be easily written up and analyzed in more depth in a future report.

5. **Total sales in each year**

   This is not part of this analysis, but it has been done and can be evaluated. It could be added to this report or a future report. In addition, this variable has been clustered. A future report could focus on the relationship between total sales, availability, and other variables.

6. **Average price per acre over 20 years**

   This has been done. It could be added to this report or a future report. In addition, this variable has been clustered. A future report could evaluate the relationship between price per acre and some of the other variables.

7. **Days on market (created date vs. closing date or sales date)**

   The days on market metric has been calculated for all of the properties.

8. **Days on market vs. parcel size**

   Because we have both days on market and parcel size, this can be done. Both parcel size and days on market have been clustered; a relationship between them could certainly be modeled, although this has not been done yet.

9. **Days on market vs. parcel value**

   As with parcel size, this could be modeled.

10. **Days on the market price vs. asking price. Was waiting justified?**

    This question is not quite clear. The relationship between the difference between asking price and sale price and days on the market could be modeled. Whether waiting was justified could be evaluated, but criteria for success would have to be established. The recession would need to be taken into account as it could skew the relationship between price difference and time on the market.

11. **Overall increase in sales value against inflation over 20 years**

    This could be examined if a strong measure of inflation over the study period is available.

12. **Overall increase in value against inflation over 20 years deflated by 1.2% holding cost, and separately 0.6% estimated property tax expense**

    As with the previous question, this could be looked at if proper inflation assumptions are made.

13. **Vacant or existing structures. Did a structure have a positive or negative effect on the value or duration of the sales process?**

    To answer this question, we would need information on which properties have existing structures.

14. **Price against selling terms (Conventional finance (typically bank loan)) vs.cash.**

    This has not yet been studied, but could be done in the future.

15. **Days on market vs. selling terms**

    This could be looked at in the future.

References